

Predicting Battery State of Health and Remaining Useful Life using Explainable Machine Learning

Adithya Prasad Shivarpatna Venkatesh
AI for Connected Industries
Module: Fundamentals of Machine Learning
Universität Ulm
Supervisor: Prof. Dr. Reinhold von Schwerin

Abstract—Lithium-ion battery State of Health (SoH) and Remaining Useful Life (RUL) cannot be measured directly in battery management systems. This paper presents a dual-domain machine learning pipeline applied to the NASA PCOE Battery Dataset (B0005–B0018), following the CRISP-DM methodology [1]. Part 1 extracts 14 physics-informed features from discharge time-series and compares Ridge, Random Forest, and XGBoost regression under a cross-battery holdout, achieving a best SoH RMSE of 0.0041 Ah. SHAP explainability analysis identifies cumulative energy throughput and internal resistance as the dominant degradation drivers. Part 2 replicates and extends Zhao et al. [2] by extracting six health factors from charging curves, forecasting their trajectories with a multi-output LSTM, and mapping predicted factors to SoH via XGBoost. Post-hoc calibration at 75th percentile (OOF residuals) reduces SoH RMSE from 0.0594 to 0.0434 Ah. Cross-domain SHAP analysis reveals that internal resistance influences capacity fade across independent measurement protocols, directly motivating a causal model as future work.

Index Terms—State of Health, Remaining Useful Life, LSTM, XGBoost, SHAP, CRISP-DM, battery prognostics, Quantile regression, explainable ML

I. BUSINESS UNDERSTANDING

Lithium-ion batteries degrade silently through repeated charge-discharge cycling. A battery management system (BMS) has no direct SoH sensor and observes only voltage, current, and temperature. Premature replacement wastes cost while delayed replacement risks unexpected failure. This poses the question “*can SoH and RUL be predicted accurately with explainability from signals already present in a BMS?*”

This project formulates two complementary regression tasks:

- 1) **Part 1 (SoH prediction + RUL):** from discharge-curve features, using current-cycle signals to estimate current-cycle capacity.
- 2) **Part 2 (HF forecasting + SoH + RUL):** from charging-curve health factors, enabling future-cycle trajectory prediction.

Cross-battery holdout. the models train on B0005, B0006, B0007 and are separately evaluated on B0018. This ensures generalization to an unseen cell setting a realistic BMS deployment scenario.

Explainability as a deliverable. SHAP values are computed for all final models to identify which features drive predictions and whether the model has learned physically

grounded relationships. Cross protocol features (Discharge and Charge domain) are compared using SHAP to assess if they are driven by the same underlying physical characteristics.

II. DATA UNDERSTANDING

A. Dataset

The NASA PCOE Battery Repository [3] provides multiple LiCoO₂ cells and four common cells (B0005 - B0018, nominal capacity 2.0 Ah) with similar test conditions are utilized for SoH prediction. They are discharged at constant 2 A to a 2.7 V cutoff at 24°C ambient, run to end-of-life (EOL) defined as 30% fade (1.4 Ah). Each cell yields approximately 168 discharge cycles and a matching sequence of charge and impedance cycles, producing approximately 200,000 raw timesteps across 636 discharge cycles.

B. Key EDA Findings

Non-monotonic degradation. Capacity curves exhibit local recovery bumps (capacity regeneration) caused by lithium redistribution during rest periods. This invalidates naive interpolation and motivates non-linear features. B0006 degrades very deeply (SoH: 0.58) while B0007 degrades slowly.

Voltage curve evolution. Discharge plateau duration shortens visibly with cycle number. The initial voltage drop at $t = 0$ increases with age which is a direct indication of rising internal resistance, justifying the use of `R_int_proxy` feature (Fig. ??).

Inter-battery variance. B0018 diverges from training batteries by up to 0.20 Ah around cycles 45-60, then partially converges mid-life. This non-stationary inter-battery divergence produces systematic early-cycle bias in cross-battery models, reported explicitly in residual analysis.

Electrochemical Impedance Spectroscopy (EIS). Electrolyte resistance (R_e : Ohm) and charge transfer resistance (R_{ct} : Ohm) offer direct evidence of linkages between resistance related features and capacity fade. (Fig. 2)

Split strategy validation. Three split strategies were compared using the same XGBoost model to quantify evaluation inflation (Table I).

TABLE I
SPLIT STRATEGY COMPARISON

Strategy	Validity
A: Random shuffle	Leakage from future cycles
B: Within-battery 50/50	validation observes train battery cycles
C: Cross-battery holdout (Group K fold)	Generalization across battery

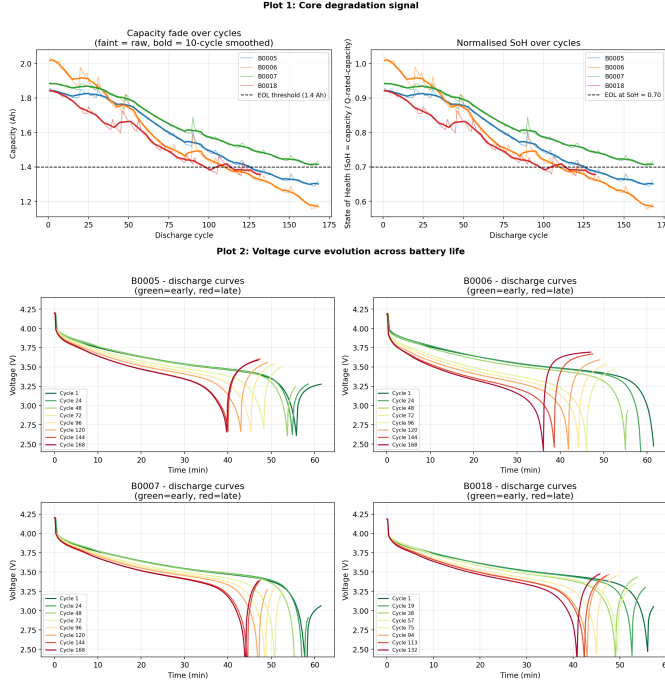


Fig. 1. EDA: capacity degradation and discharge curve evolution across battery life

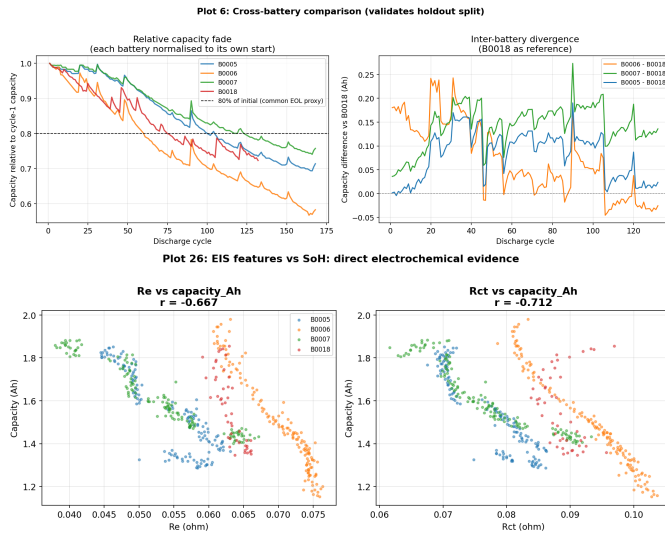


Fig. 2. EDA: Cross-battery Variance and capacity by EIS fields

III. DATA PREPARATION

A. Part 1: Discharge Feature Engineering

Each discharge cycle (300 time steps) is collapsed to 14 scalar features divided into two categories:

Statistical features: discharge duration, voltage mean/std/slope, voltage at 80% of discharge, temperature rise.

Physics-informed features: internal resistance proxy ($R_{int} \approx dV/dI$ at cycle start), cumulative charge (Q_{cum}), energy throughput (E_{Wh}), temperature-compensated capacity (Q_{norm}), incremental capacity peak height and voltage (dQ/dV). Smoothened R_{int} and flag to capture initial bathtub curve where internal resistance drops.

Rolling features (`rolling_mean_5`, `rolling_std_5`) are computed with a `shift(1)` offset to prevent target leakage the rolling mean for cycle n uses only cycles $n - 5$ to $n - 1$.

The Correlation between features and capacity reveals possible high value features. Autocorrelation concerns are handled internally by ensemble methods such as XGBoost. (Fig. 3). All features pass through a `sklearn` Pipeline (`StandardScaler` → `model`) fit exclusively on training batteries, preventing leakage.

B. Part 2: Charge Health Factor Extraction

Following Zhao et al. [2], six health factors (HF1–HF6) are extracted from the CC - CV charging curve per cycle. Spearman correlation (insted of paper's MIC analysis) selects HF1, HF2, HF3, HF6 (Table II) as inputs to the LSTM.

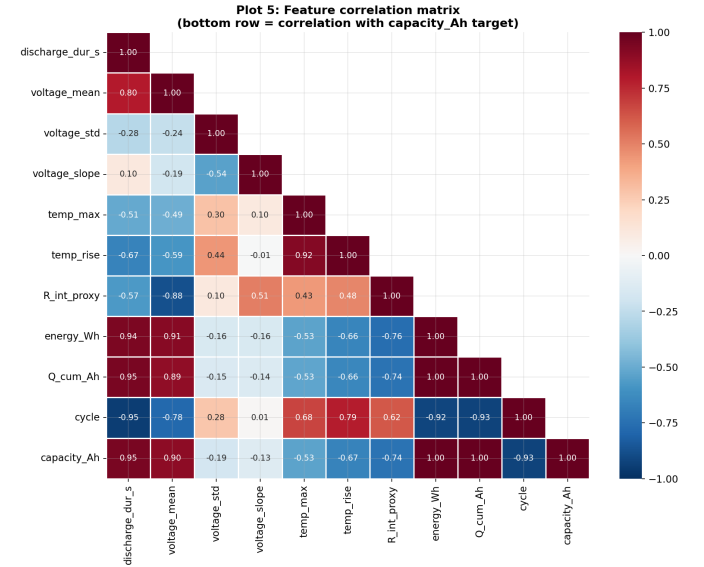


Fig. 3. Correlation Heatmap: Features and Capacity

TABLE II
HEALTH FACTOR SELECTION VIA SPEARMAN CORRELATION

HF	Description	Spearman r	Selected
HF1	CC charge time 3.9→4.2 V	0.935	✓
HF2	Voltage rise within 600 s	-0.919	✓
HF3	CV current drop within 900 s	0.598	✓
HF4	Exp. fit floor (I_∞)	-0.172	×
HF5	Exp. fit amplitude	0.030	×
HF6	RC time constant	-0.871	✓

IV. MODELING

A. Part 1: Discharge Domain Models

Three models are trained and evaluated with GroupKFold cross-validation (3 folds, grouped by battery ID, train on B0005, B0006, B0007 test on B0018):

- **Ridge Regression** (baseline): L_2 -regularized linear model ($\alpha = 1.0$). Coefficients are directly interpretable, and features proportionally influence the target variable. SHAP via `LinearExplainer` provides the feature rank and direction.
- **Random Forest**: 200 trees, `min_samples_leaf=2`.
- **XGBoost**: 300 estimators. Tree based SHAP provides the feature rank and also for feature engineering statistical features (Fig. 4).
- The hyperparameters are tuned using `GridSearchCV` with `GroupKFold` validation.



Fig. 4. Ridge and XGBoost SHAP analysis.

B. Part 2: LSTM + XGBoost Pipeline

The two-stage pipeline follows Zhao et al. [2] with XGBoost replacing GPR:

- **Stage 1: Multi-output LSTM**: A 2-layer LSTM (hidden size 32, dropout 0.3, Adam $\text{lr}=0.0005$) predicts all four

HF values at cycle $n + 1$ from a sliding window of the last 10 cycles. Cross battery validation split (train on 2 batteries and validate on the remaining one). Dropout and gradient clipping prevent overfitting. Early stopping is applied and mean `MAX_EPOCH` across validation is used for full model training (Fig. 5).

- **One-step vs. rollout**: One-step predictions (true history window) demonstrate captured trend while auto-regressive rollout pipeline provides a deployment simulation. Rollout suffers from compounding variance (Fig. 6).
- **Stage 2: XGBoost on predicted HFs**: XGBoost trained on HF1, HF2, HF3, HF6 from training batteries maps predicted HF values to SoH. The cycle index is explicitly excluded to prevent target leakage.
- **Post-hoc calibration**: Raw predictions exhibit systematic under-prediction on B0018 due to inter-battery capacity offset and LSTM error propagation. The OOF mean shift, Q75 / Q90 residual shift, and XGBoost quantile regression were explored to mitigate bias. The OOF mean shift worsens the performance as training batteries show bias in opposite-direction. 75th quantile residual shift achieves lowest bias. Native quantile XGBoost underperforms relative to post-hoc shift although provides a robust path for deployment pipeline (Fig. 7).

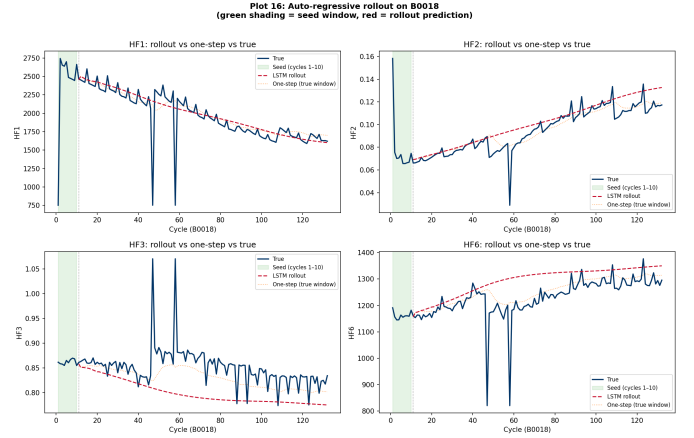


Fig. 5. LSTM predictions across horizon.

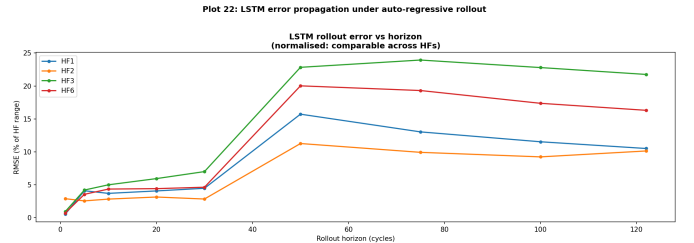


Fig. 6. LSTM error propagation under auto-regressive rollout. All HFs converge to 15-22% of their operating range by horizon 50, driven by anomalous B0018 cycles 45-60.

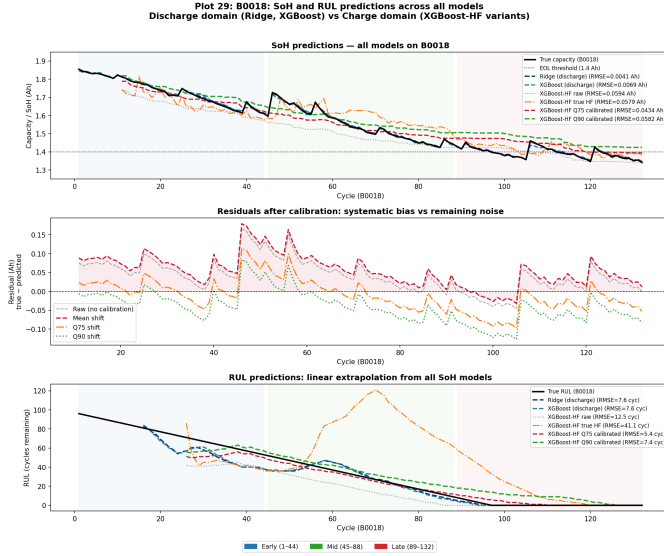


Fig. 7. SoH (with post-hoc calibration) and RUL predictions across all models on B0018.

V. EVALUATION

A. SoH Prediction

Table III summarizes SoH RMSE across all models on B0018. Discharge-domain models outperform charge-domain models. This is attributed to the fact discharge features are measured during the same event being predicted. Q75 calibration reduces charge-domain RMSE by 26% and demonstrates the best performance.

TABLE III
SoH RMSE ON B0018 HOLDOUT

Model	RMSE (Ah)	Domain
Ridge regression	0.0041	Discharge
XGBoost (discharge)	0.0069	Discharge
Paper GPR+LSTM [2]	0.0150	Charge (within-bat.)
XGBoost-HF Q75 calibrated	0.0434	Charge (cross-bat.)
XGBoost-HF true HF	0.0579	Charge (cross-bat.)
XGBoost-HF raw	0.0594	Charge (cross-bat.)

Residual analysis on discharge model.

- Ridge shows a systematic sign pattern. early cycles are diverse while mid and late cycles trend negative, meaning Ridge over-predicts as the battery ages. (Fig. 8)
- Random Forest and XGBoost under-predicts capacity as degradation accelerates. This is the fingerprint of ensemble overfitting due to recovery bump patterns of training batteries, validating the inter-battery offset seen in EDA.
- Across all three predicted-vs-actual plots, points cluster tightly on the diagonal throughout, confirming the models remain reliable near EOL where accuracy matters most for replacement decisions, despite their mid-life residual structure.

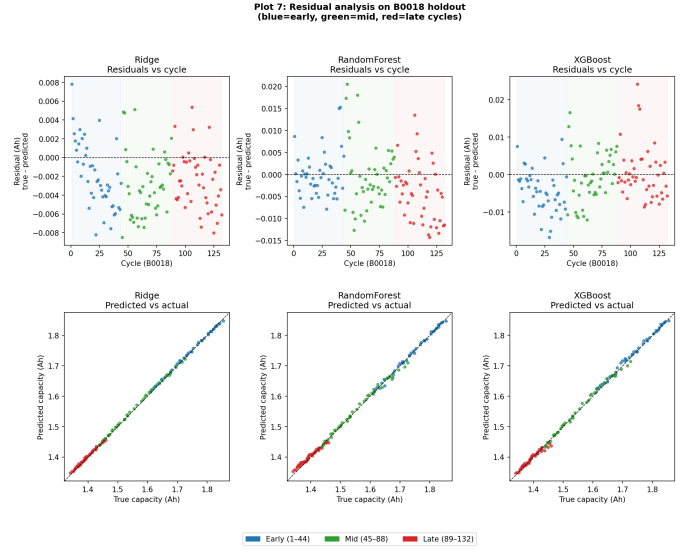


Fig. 8. Residual Analysis on Discharge models.

B. RUL Prediction

Linear extrapolation from each model's SoH trajectory yields the RUL estimates in Table IV. Discharge model (XGBoost) achieves lower rmse relative to raw LSTM+XGBoost model. Post-calibration with 75th percentile of OOF residuals achieves the best approximation of RUL curve across the life-cycle of battery (Fig. 7).

TABLE IV
RUL RMSE ON B0018 (LINEAR EXTRAPOLATION)

Model	RMSE (cycles)	MAE (cycles)
XGBoost (discharge)	7.56	5.03
Ridge (discharge)	7.64	4.94
Paper GPR+LSTM [2]	5.54	0.58
XGBoost-HF raw	12.47	9.47
XGBoost-HF Q75 calibrated	5.44	3.32
XGBoost-HF Q90 calibrated	7.45	6.38

C. Explainability: Cross-Domain SHAP Convergence

SHAP analysis on both domain models reveals convergent evidence for a two-pathway degradation model (Fig. 9):

Discharge domain: `energy_Wh` and `Q_cum_Ah` dominate. These are direct capacity proxies, cumulative energy and charge delivered per cycle. `R_int_proxy` lower due to multicollinearity with the throughput features, but retains physical significance as the resistance mechanism underlying their decline.

Charge domain: `HF6_rolling_mean5` (RC time constant trend) and `HF1` (CC charge time) dominate. Both measure how internal resistance affects the charging process as higher resistance lengthens CC charging and increases the RC time constant.

Convergence interpretation: As batteries age, internal resistance increases. From the discharge side, this reduces

energy deliverable per cycle ($\text{energy_Wh} \downarrow$). From the charge side, this lengthens the time constant of CV current decay (HF6 $\uparrow$) and slows CC charging (HF1 $\uparrow$). Both pathways measure the same electrochemical mechanism from independent experimental protocols.

Degradation rate correlation reinforces this: HF1 evolution rate correlates with capacity degradation rate at $r = 0.997$ ($p = 0.003$) across all four batteries indicating a near-perfect cross-domain link. HF6 shows consistent direction ($r = -0.882$) but does not reach significance at $n = 4$.

EIS corroboration: Impedance spectroscopy measurements show R_{ct} (Spearman $r = -0.70$) and R_e ($r = -0.64$) negatively correlated with capacity, providing independent electrochemical confirmation that resistance increase accompanies capacity fade [4]. EIS features are omitted due to sparse availability (48 of 132 B0018 cycles).

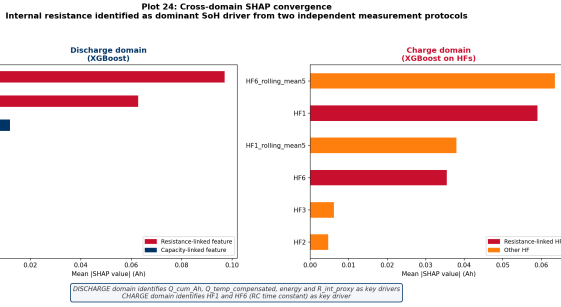


Fig. 9. Cross-domain SHAP convergence. Discharge domain (left) identifies cumulative throughput features; charge domain (right) identifies RC time constant and CC charge time. Both implicate internal resistance as the primary SoH driver.

VI. DEPLOYMENT & FUTURE WORK

A. Deployment Considerations

The discharge-domain ridge model (RMSE 0.0041 Ah, RUL RMSE 7.6 cycles) is the recommended deployment model for BMS applications where discharge telemetry is available. It is interpretable, well-calibrated, and achieves competitive RUL accuracy without requiring sequence modeling.

The charge-domain Q75 pipeline is preferable when *future* SoH is needed from standardized charging sessions which is a common scenario for fleet BMS where discharge conditions vary. The raw LSTM+XGBoost RUL predictions is operationally justified under conditions where under-predicting SoH (conservative) is safer than over-predicting, as premature replacement is costly but unexpected failure could be dangerous.

A recalibration frequency of 30-40 cycles is recommended for the LSTM rollout, based on the error propagation analysis showing RMSE stabilization at horizon 50.

B. Limitations

Results are evaluated on a single test battery (B0018). The cross-battery generalization gap is partially explained by B0018's anomalous cycles 45-60 (confirmed experimental interruption) and its lower starting capacity relative to training

batteries. A larger battery cohort would improve both model training and evaluation robustness.

The charge-domain XGBoost suffers systematic inter-battery bias that OOF calibration cannot correct because training and test residuals have opposite signs. The health-factor forecast is seeded from only the first 10 true cycles and then self-feeds for the remaining 120+ cycles. For longer cycles LSTM prediction errors are cumulative.

C. Causal Roadmap

The cross-domain SHAP convergence and degradation rate correlations motivate a DAG-based causal model as possible follow up:

$$\text{Aging} \rightarrow R_{\text{int}} \uparrow \rightarrow \begin{cases} E_{\text{Wh}} \downarrow & (\text{discharge}) \\ \text{HF6} \uparrow, \text{HF1} \uparrow & (\text{charge}) \end{cases} \rightarrow \text{SoH} \downarrow \quad (1)$$

Using DoWhy, the intervention question becomes: *if R_{int} is held constant (e.g., via temperature control or formation cycling), what is the counterfactual effect on capacity fade rate?* This requires the present correlation model as a necessary precursor, establishing which variables carry signal before asking which are causal.

VII. CONCLUSION

This work demonstrates that battery SoH and RUL can be predicted accurately with explainability from both discharge and charging signals using tabular ML methods. Discharge-domain Ridge regression achieves RMSE 0.0041 Ah outperforming more complex architectures under a stricter evaluation protocol.

The LSTM+XGBoost charge-domain pipeline replicates and extends Zhao et al. [2], replacing GPR with a calibrated XGBoost and adding SHAP explainability absent from the original paper. Cross-domain SHAP analysis reveals that internal resistance, measured as cumulative throughput decline from discharge curves and RC time constant growth from charging curves, is the convergent driver of SoH degradation across independent experimental protocols. This convergence constitutes the primary finding and directly motivates a causal model as future work.

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